

GreenVision.AI

AI-Powered Environmental Intelligence
for Cities, Industries & People

Turning environmental data into green action

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Executive Summary

GreenVision.AI is an AI-powered environmental intelligence platform that transforms raw environmental data into actionable green decisions. It provides three distinct workspaces: Municipal (city planning), Industrial (compliance assessment), and Citizen (community engagement), each tailored to a different stakeholder.

The system combines machine learning models (YOLOv8 for tree detection, U-Net for canopy segmentation, scene classification) with real-time weather, air quality, soil, and elevation data from Open-Meteo. A conversational AI assistant provides location-conditioned species recommendations with cost estimates.

Verified results from live analysis: 7.92% canopy coverage detected, 156 trees identified, 1026 additional trees needed for 60% green cover target, estimated carbon offset of 2.2 t/yr at approximately Rs. 35,000 investment.

Key Metrics

Canopy Coverage: 7.92% | Trees Detected: 156 | Trees Needed: 1026 | Carbon Offset: 2.2 t/yr | Test Suite: 35/35

Problem Statement

Indian cities are experiencing rapid loss of green cover. According to the India State of Forest Report (ISFR 2023), urban areas lose 1.5 to 3 percent of tree cover per decade due to infrastructure expansion, urbanization, and insufficient plantation planning.

Key Challenges

- No real-time tool exists to measure urban green cover from aerial or satellite imagery
- Municipalities plan plantations without data-driven priority mapping
- Industrial sites lack systematic green buffer compliance assessment
- Citizens have no way to assess or act on local environmental data
- Existing GIS tools require specialized expertise and are not accessible

Why It Matters

Green cover directly impacts urban heat island effects, air quality, stormwater management, and citizen well-being. Without accurate, accessible environmental intelligence, cities cannot make informed decisions about where to plant, how many trees to plant, or which species are most appropriate.

Solution Architecture

GreenVision.AI is built as a full-stack web application with a React frontend, Flask backend, and ML pipeline that processes environmental imagery and data.

Technology Stack

- Frontend: React 19, Vite, TailwindCSS, React-Leaflet, jsPDF
- Backend: Flask (Python 3.11), REST API
- ML Pipeline: PyTorch 2.2, YOLOv8 (tree detection), U-Net (canopy segmentation)
- External Data: Open-Meteo API (weather, AQI, soil, elevation)
- Geocoding: Nominatim (OpenStreetMap) with rate limiting compliance
- Storage: Flat JSON files (demo-grade), localStorage for user identity

System Flow

1. User uploads imagery (aerial, satellite, or ground-level photo). 2. Scene classifier determines environment type. 3. U-Net segments canopy coverage. 4. YOLOv8 detects individual trees. 5. Location context fetched via Open-Meteo. 6. Species engine recommends trees. 7. Results presented with MEASURED/ESTIMATED/UNAVAILABLE labels.

Municipal Workspace

The Municipal workspace is designed for city planners and urban forestry managers. It provides a comprehensive green cover diagnosis for any uploaded aerial or satellite image.

Key Features

- Upload aerial or satellite imagery of any city area
- AI-powered canopy segmentation and tree detection
- Green cover percentage calculation against 60 percent target
- Planting gap analysis: how many trees are needed
- GIS decision-support map with color-coded priority zones
- Carbon sequestration and oxygen production estimates
- One-click executive PDF report generation

Workflow

The municipal user uploads an image, the system runs the full ML pipeline, and produces a comprehensive diagnosis. The priority map highlights areas with lowest canopy coverage for immediate intervention.

Industrial Workspace

The Industrial workspace provides Environmental Compliance Assessment for factory and industrial site managers. It evaluates green buffer zones around industrial perimeters.

Key Features

- Site green buffer assessment and compliance status
- Compliance tracking: Below Threshold, Needs Improvement, or Acceptable
- Buffer zone planning with pollution-tolerant species recommendations
- Investment estimation for green buffer development
- ESG (Environmental, Social, Governance) reporting metrics
- Maintenance planning and schedule generation

Compliance Framework

The system evaluates the current green buffer percentage against regulatory thresholds. Sites below threshold receive urgent remediation recommendations. The system honestly notes that it assesses green coverage, not actual pollution levels.

Citizen Workspace

The Citizen workspace provides a personalized, location-aware environmental experience. Each citizen gets a dashboard tailored to their specific neighborhood conditions.

Key Features

- Automatic location detection via browser geolocation
- Interactive Leaflet map showing the user area
- Real-time weather, AQI, and soil data for the location
- AI chatbot for natural conversation about local trees
- Green Points system for community contributions
- Community leaderboard and challenges

Personalization

The Citizen workspace greets users by name stored locally, provides a time-appropriate greeting, and offers environmental data specific to their exact location. The AI chatbot maintains session memory and provides follow-up suggestions based on conversation context.

AI Chatbot

The GreenVision AI assistant is not a simple keyword matcher. It is a context-aware environmental advisor that maintains conversation state and provides personalized, location-conditioned recommendations.

Conversational Capabilities

- Session memory: remembers user name and prior recommendations
- Name-aware greetings with user first name
- Follow-up context: tracks last intent, species shown, cost discussed
- Species exclusion: when asked for alternatives, excludes previously shown species
- Cost follow-ups: if user asks about budget, provides cost estimate
- Honest fallback when unknown species are requested

Example Conversation

User says hi and gives name. Bot greets by name. User asks what trees are good for area. Bot ranks species by location score. User asks for another option. Bot re-ranks excluding shown species. User asks how many trees needed. Bot provides count to reach 60 percent target.

Species Engine

Location-conditioned species recommendations using real environmental data. No hardcoded numbers for weather, AQI, or soil values.

Data Sources

- Temperature, humidity, wind speed from Open-Meteo current weather
- Air Quality Index including PM2.5 and PM10 from Open-Meteo AQI
- Soil pH, sand, silt, clay fractions, organic carbon from SoilGrids
- Elevation above sea level from Open-Meteo

16-Species Database

Scoring weights: temperature 15pts, rainfall 15pts, AQI 15pts, soil pH 10pts, elevation 10pts, air quality 10pts, urban tolerance 15pts, hardness 10pts. Total 100pts.

- Species include: Neem, Pongamia, Peepal, Banyan, Arjuna, Cassia, Gulmohar, Mango, Tamarind, Jackfruit, Mahogany, Bamboo, Teak, Sal, Rain Tree, Amaltas

Example: Bengaluru

For tropical humid Bengaluru: Neem scores 82/100 (tropical climate, AQI tolerant). Pongamia scores 75/100 (pollution tolerant, versatile soil). Peepal scores 72/100 (urban tolerant, high oxygen).

Organization Ecosystem

Community-driven contributions with organization-level aggregation. From individual action to collective impact.

Contribution System

- 6 org types: Individual, Company, College, School, NGO, Community
- Organization name stored with each contribution in backend
- Backend aggregates by org: unique participants, total contributions, total points
- 4 action types: Plant Tree (100pts), Maintain (50pts), Community (150pts), Green Space (75pts)

Leaderboard

- Individual rankings with podium display and AnimatedNumber counters
- Organization tabs: Companies, Colleges, NGOs, Communities, Schools
- Real aggregation from contributions.json via /api/leaderboard endpoint
- Green Challenges for ongoing community engagement

Important: Green Points measure participation activity, not CO2 or O2 production. Environmental estimates come from ML analysis, not contribution scores.

ML Pipeline

Three complementary ML models process uploaded imagery in parallel.

Model 1: YOLOv8n - Tree Detection

Ultralytics YOLOv8n pretrained on COCO dataset. Detects potted plants and trees. Runs on CPU. Uses non-max suppression with confidence threshold above 0.25.

Model 2: U-Net - Canopy Segmentation

Encoder-decoder architecture with skip connections. EfficientNet-B0 backbone. Sigmoid output for binary vegetation mask. Pixel-level canopy ratio equals area percentage.

Model 3: Scene Classifier

EfficientNet-B0 trained on 20 scene categories: agricultural, airplane, airport, beach, bridge, city, desert, forest, grassland, harbor, mountain, parking, residential, river, runway, ship, snowfield, sparse residential, stadium, viaduct.

Pipeline Stages

1. Image preprocessing (224x224 resize, normalize).
2. Scene classification for environment context.
3. U-Net segmentation for canopy pixels.
4. YOLOv8 detection for tree count.
5. Aggregate metrics (area, density, priority).
6. Location context via Open-Meteo.
7. Species recommendations conditioned on all data.

Verified Results

All numbers from live analysis of low_canopy_urban_aerial.jpg (August 2026).

Image Analysis

- File: low_canopy_urban_aerial.jpg
- Scene: street (98.12% confidence)
- Canopy Coverage: 7.92% (MEASURED)
- Trees Detected: 156 (MEASURED)
- Priority: High (canopy below 25%)

Environmental Estimates

- Carbon Sequestration: 3.43 t/yr (ESTIMATED)
- Oxygen Production: 18.41 t/yr (ESTIMATED)
- Analysis Area: 0.5191 hectares
- Tree Density: Sparse (0.167)

Gap Analysis

- Trees needed to reach 60% canopy: 1,026
- Estimated budget: Rs. 2,77,020 (Rs. 270 per tree)

Data Honesty Labels

MEASURED: canopy percentage, tree count (direct ML output). ESTIMATED: carbon offset, oxygen (based on tree models). UNAVAILABLE: pollutant levels, disease detection, soil contamination.

Impact Assessment

Environmental Impact

- Urban heat island reduction through data-driven tree placement
- Air quality improvement tracking and prediction
- Stormwater management through canopy analysis
- Biodiversity support via native species recommendations

Social Impact

- Community engagement through Green Points and leaderboards
- Environmental awareness via accessible data visualization
- Citizen participation in urban greening decisions
- Educational value through species information

Economic Impact

- Reduced plantation costs through targeted planting
- ESG compliance support for industrial sites
- Municipal budget optimization for green infrastructure
- Property value enhancement through strategic greening

Feasibility and Viability

Technical Feasibility

- Built on open-source stack: React, Flask, PyTorch
- Models run on CPU, no GPU dependency
- Open-Meteo API provides free environmental data
- Standard web hosting sufficient for deployment

Economic Viability

- No licensing costs for core technology
- Minimal hosting expenses for static frontend and Flask backend
- Scalable architecture from ward to city to state to national level
- Revenue potential through premium analytics features

Operational Viability

- Phased deployment starting with single ward or city
- Integration with existing municipal GIS systems
- Minimal training required for end users
- Mobile-responsive design for field use

Acknowledged Limitations

Demo-grade: flat JSON storage, no authentication, no pollution model, species coverage focused on Indian tropical/subtropical regions. These are documented and planned for improvement.

References and Future Work

References

- Smart Cities Mission, MoHUA, Government of India
- ISRO Bhuvan Geoportal for satellite imagery
- India State of Forest Report (ISFR 2023)
- Open-Meteo API for open-source weather data
- YOLOv8 and Ultralytics for object detection framework
- React frontend framework
- Flask Python web framework
- SoilGrids for global soil information

Future Roadmap

- Pollutant dispersion model for industrial mode
- Historical trend analysis with multi-temporal comparison
- User accounts with verified identity
- Mobile app for field verification
- Integration with municipal GIS databases
- Multi-language support for wider accessibility